

AR 503: Mechatronics

Course Code	AR 503
Course Name	Mechatronics
L-T-P-C	3-0-0-3
Prerequisites	Consent of faculty advisor
Intended for	UG, PG and PhD
Mutual Exclusion	None
Approval	50th BoA (Jul 2023)

Course Contents

- **Introduction to Mechatronics:** Introduction, Examples of Mechatronic systems, Electric circuits and components, Review of fundamentals of Electronics and its applications, Number systems: binary, hexadecimal and Review of C programming, CNC machines and Industrial Robots. (4 hours)
- **Mechatronics elements:** Sensors and transducers, Displacement, Position & Proximity Sensors, Force, Fluid pressure, Liquid flow sensors, temperature, light sensor, Acceleration and, Vibration measurement, Performance terminology of sensors, Semiconductor sensors and micro-electromechanical systems (MEMS). (8 hours)
- **Microprocessors, microcontrollers, and Closed-loop controllers:** Digital circuits, Microprocessors, Microcontrollers, Programming of Microcontrollers, P, I, PID Controllers, Digital Controllers, Program Logic Controllers, Input/output & Communication systems, Fault finding. (4 hours) Signal conditioning: Introduction to signal processing, Op-Amp as a signal conditioner, Analog to Digital Converter, and Digital to Analog Converter. (4 hours)
- **Actuators, Drives, and mechanisms:** Stepper motors, Actuators, Motor sizing, Power transmission: gears (rack and pinion, spur, planetary, worm, bevel, crown, harmonic) and belt drives; Ball screws, linear motion bearings, cams, systems controlled by camshafts, electronic cams, indexing mechanisms, tool magazines, and transfer systems. Torque, speed, and power equations, efficiency, and inertia. (6 hours)
- **Hydraulic and Pneumatic system:** Flow, pressure, direction control valves, actuators, and supporting elements, hydraulic power packs, pumps, and design of hydraulic circuits. Pneumatic system production, distribution and conditioning of compressed air, system components and graphic representations, design of systems. (6 hours)
- **Modelling and system response:** Mechanical, Electrical, Fluid system modelling, Dynamic response, Transfer function and frequency response. (6 hours)
- **Final project on mechatronics:** Group project towards design and fabrication of functional Mechatronics systems. (4 hours)
- **Laboratory/practical/tutorial Modules:** Microprocessors, microcontrollers, and Closed-loop controllers, Signal conditioning, Actuators, Drives, and mechanisms, Hydraulic and Pneumatic system, Modelling and system response.

Text Books

1. Bolton, William, **Mechatronics: electronic control systems in mechanical and electrical engineering**, Pearson Education.
2. Mahalik, Nitaigour, Premchand, **Mechatronics**, Tata McGraw-Hill.
3. Bishop, Robert H, **Mechatronics: an introduction**, CRC Press.
4. Devdas Shetty and Richard A. Kolk, **Mechatronics system**, Cengage Learning.
5. G. Hegde, **Mechatronics**, Jones and Bartlett.

References

1. G.W. Kurtz, J.K. Schueller, P.W. Claar II, **Machine design for mobile and industrial applications**, SAE.
2. HMT Ltd. **Mechatronics**, Tata McGraw-Hill.
3. T.O. Boucher, **Computer automation in manufacturing - an Introduction**, Chapman and Hall.
4. Kevin Lynch, Nicholas Marchuk, Matthew Elwin, **Embedded Computing and Mechatronics with the PIC32 Microcontroller**, Newnes.
5. R. Iserman, **Mechatronic Systems: Fundamentals**, Springer.
6. Musa Jouaneh, **Fundamentals of Mechatronics**, Cengage Learning.